HYDROGELS: EVALUATING THE VIABILITY OF HYDROGELS IN SALINE CONDITIONS

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Abstract:
The recent galamsey crisis in Ghana has left water bodies polluted with chemical contaminants, depriving farmers of clean irrigation sources. This study evaluates the effects of dissolved salts, specifically NaCl and $MgCl_2$, on the water absorption and release kinetics of biodegradable hydrogels composed of starch, polyvinyl alcohol (PVA), and borax, as a potential solution to this issue. Hydrogels were synthesized in a 2:1:1 ratio, dried at 40°C, and subsequently immersed in saline solutions at 5%, 10%, and 15% (w/v) concentrations, with distilled water as a control. Results revealed that distilled water exhibited the highest absorption and release rates. Among saline solutions, 15% NaCl had higher absorption rates than 5% and 10%, while 15% $MgCl_2$ performed better than lower concentrations. Statistical analyses, including One-Way ANOVA and Tukey's HSD tests, confirmed significant variations in absorption and release rates across different concentrations and salt types. However, no consistent trend was observed between increasing ionic concentration and hydrogel performance, suggesting complex interactions between salt type, ionic strength, and the polymer network. These findings underscore the need for hydrogel formulations optimized for ionic environments to ensure effective water management in agriculture, particularly in regions with saline or polluted water sources.
Keywords: Sodium Chloride, Hydrogel, Water Absorption, Water Release, Magnesium chloride, Agriculture, Illegal mining, T-test, Normality- test, p-Value, ANOVA test, Post-hoc analysis, Tukey HSD

Introduction

Hydrogels, polymeric materials characterized by their high-water absorption and retention capabilities, are increasingly leveraged in agricultural applications to mitigate water scarcity. Despite their widespread use, the behavior of hydrogels in contaminated water environments—where dissolved ionic impurities such as salts disrupt their structure and function—remains insufficiently understood. This study addresses this critical knowledge gap by evaluating the impact of dissolved salts on the water absorption and release kinetics of hydrogels. From secondary research, we found sodium and magnesium were common salts present in water. Hydrogels underwent controlled drying and were subsequently immersed in NaCl and $MgCl_2$ solutions at 5%, 10%, and 15% w/v concentrations. Experiments were performed under non-soil conditions to isolate ionic effects without the confounding influences of soil matrices [3].

Experimental Design

Our investigation began into which salts to use for our experiments. After secondary research, we found that sodium, magnesium, and calcium were the most common salts found in water bodies; however, due to resource limitations, we could obtain only sodium and magnesium. From research, the metrics which decide which hydrogel is effective or not are water release rates, which check how much water a hydrogel can release and water absorption rates, which evaluate how much water a hydrogel can absorb from the environment after depleting its stores. We hypothesize that introducing ionic solutes disrupts the hydrogel's hydrogen bonding network and osmotic gradient, leading to reduced water uptake and slower release rates. We also hypothesize that increasing concentration of salt will reduce water release and water absorption rates since high concentrations may disrupt hydrogel structures.

Materials and Methodology

Materials

The materials used in this study included starch (15% w/v), polyvinyl alcohol (PVA) (5% w/v), borax (10% w/v), sodium chloride (NaCl) (5%, 10%, and 15% w/v), magnesium chloride ($MgCl_2$) (5%, 10%, and 15% w/v), and distilled water. Equipment comprised 50 ml beakers, standardized molds, an incubator maintained at 40°C, a weighing scale, a timer, and tissue paper as blotting paper.

Hydrogel Synthesis:

Hydrogels were synthesized using a biodegradable formulation comprising 15% starch, 5% polyvinyl alcohol (PVA), and 10% borax by weight. Starch was selected for its superior water absorption capabilities, PVA for its film-forming properties, and borax for its role as a crosslinking agent. The components were mixed thoroughly in a 2:1:1 ratio by volume (100 ml starch, 50 ml PVA, 50 ml borax) to form a homogenous solution. The mixture was poured into standardized molds and dried in an incubator at a controlled temperature (e.g., 40°C) for 24 hours to ensure consistent crosslinking and structural integrity.

Absorption Test Procedure

To assess the impact of ionic concentration on water absorption, six hydrogel samples (2 g each) were prepared and immersed in aqueous solutions of NaCl and $MgCl_2$ at three concentrations: 5%, 10%, and 15% (w/v), along with a control sample in distilled water. Each solution was prepared with a fixed volume of 15 ml.

- The hydrogels were placed in separate beakers containing the respective solutions.
- Weight measurements were taken at 5-minute intervals over a total duration of 60 minutes (12 measurements per sample).
- Each hydrogel was removed, gently blotted to remove surface moisture, and immediately weighed using a precision analytical balance.

Release Test Procedure

After the absorption phase, the hydrogels were subjected to controlled drying to observe water release rates.

- The previously absorbed hydrogels were placed in an incubator maintained at 40°C to facilitate water release.
- Weight measurements were taken every 15 minutes over 4 hours (16 intervals), followed by three additional measurements at hourly intervals.
- The process was designed to capture both the short-term and longer-term release rates

Analysis & Tests

In investigating our research question, we settled on using a One-Way ANOVA test to check whether there was a difference in water release and absorption rates between different salts. The One-Way Anova was chosen since we had one independent factor, the varying salt concentration, which influenced the water release and absorption of the hydrogel formulation we were testing. We came to this conclusion after validating the normality of our data and taking into consideration that our analysis was going to involve comparison between more than two independent groups with the assumption of equal variance. We obtained normality from the KS-Limiting form test, where we had a p-value greater than 0.05, indicating that our data came from a normally distributed population, validating our need to use a parametric hypothesis test. We then proceeded to test for equality of variance of our data with the Bartletts test, given that our data was normally distributed, and we obtained a p-value greater than 0.05, further validating our assumptions. After performing our One-Way ANOVA test, we obtained a p-value that was less than 0.05, indicating that at least one of the groups was statistically different from the rest, probing us to conduct a post-hoc test to further our investigation.

Our focus was on identifying the specific differences between the groups and so we performed a post-hoc Tukey's Honestly Significant Difference test since it has the greatest specificity when it comes to pairwise comparisons out of the various parametric post-hoc tests available such as Bonferroni, Scheffe test, etc. This is mainly due to its moderate level of conservativeness, which allows it to better detect differences between groups in a robust manner.

Additionally, we performed an unpaired t-test to determine whether there was a statistical difference between the water release and absorption rates of the distilled water hydrogels used as controls for the NaCl and the $MgCl_2$ samples. This was valid as we were comparing two unpaired groups. We had a p-value greater than 0.05, meaning there was no difference.

Results

Water Absorption Rates for NaCl

From Fig 1.1 and bar graph, Fig 1.5, distilled water had the highest slope and water holding capacity as compared to the rest of the salts. This was verified by the Tukey's HSD test which revealed statistical difference($p<0.05$) between distilled water and the various concentrations, 5%, 10%, and 15% concentrations.

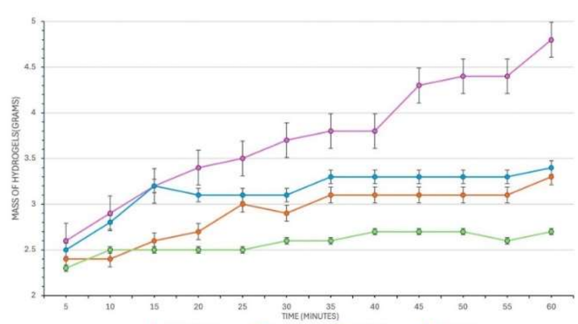

Fig 1.1 Line graph showing the water absorption for the varying concentrations of NaCl

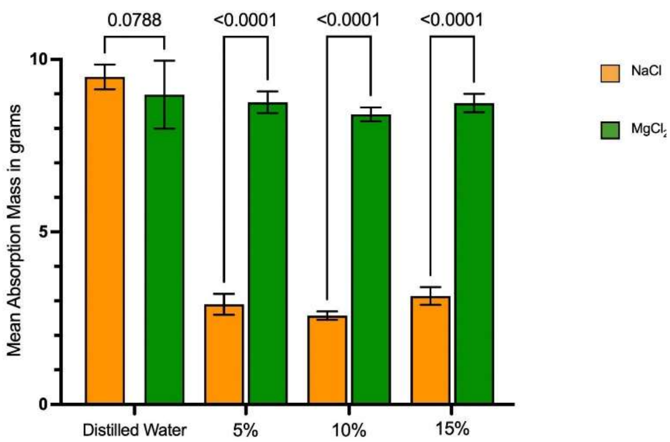

Fig 1.5 Graph of the mean absorption mass of the hydrolgels treated with variying Nacl and $MgCl_2$ Concentrations

Between the salts, there was a difference in mean absorption rates between only 10% and 15%. However, there was no difference between the release rates of 5% and 10% and 5% and 15% with p-values> 0.05, indicating similar release rates.

Water Release Rates for NaCl

There was a difference in mean release rates between distilled and the various concentrations, with a p-value<0.05. From the line graph Fig 1., distilled water had the highest release rate, followed by 15% NaCl, 5% NaCl and 10% NaCl.

Between all pairwise comparisons of the salts, 5% and 10%, 5% and 15% and 10% and 15%, there was a difference in the mean release rates since p-value<0.05, indicating they had different release rates.

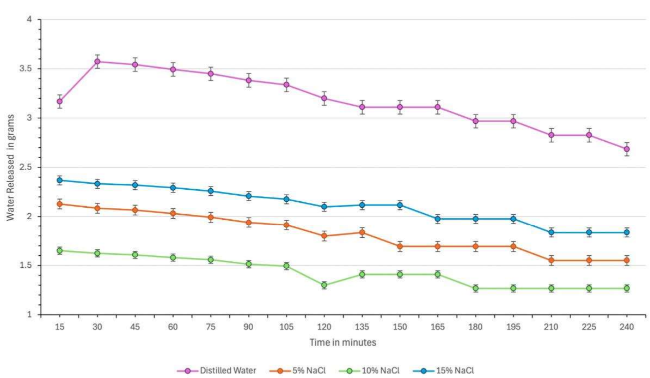

Fig 1.2 Line graph showing the water release for varying concentrations of NaCl

Water Absorption Rates for $MgCl_2$

There was a difference in mean release rates between distilled and the various concentrations, with a p-value<0.05. From Fig 1.3, it can be observed distilled water had the highest absorption rates followed by 5% and 15% concentrations. Fig 1.5 also illustrates that distilled water had the highest water holding capacity. Evaluating pairwise comparisons between the salts, only 5% and 15% $MgCl_2$ showed no difference in their absorption rates with p-values greater than 0.05. However, the remaining comparisons, 5% and 10% and 10% and 15%, showed statistical significance indicating there is a difference in their mean water absorption rates.

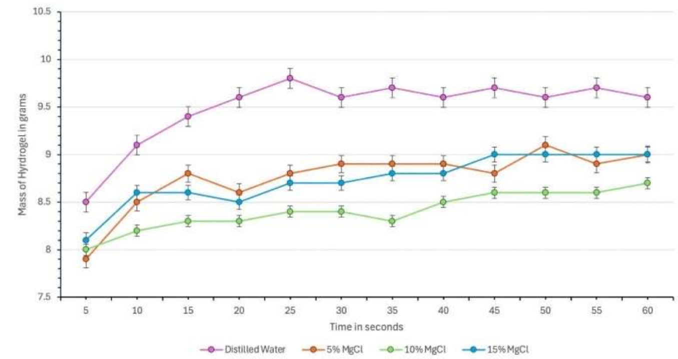

Fig 1.3 Line graph showing the water absorption for the varying concentrations for $MgCl_2$

Water Release Rates for $MgCl_2$

Fig1.4 showed that distilled water had the highest water release rates compared to the other salts. The post-hoc analysis verified it with a difference in mean water release rates between distilled water and the three concentrations with p-values less than 0.05.

However, all pairwise comparisons between the salts revealed no statistical significance with p-value<0.05

This means that the water release for the $MgCl_2$ salts were the same, regardless of the concentration

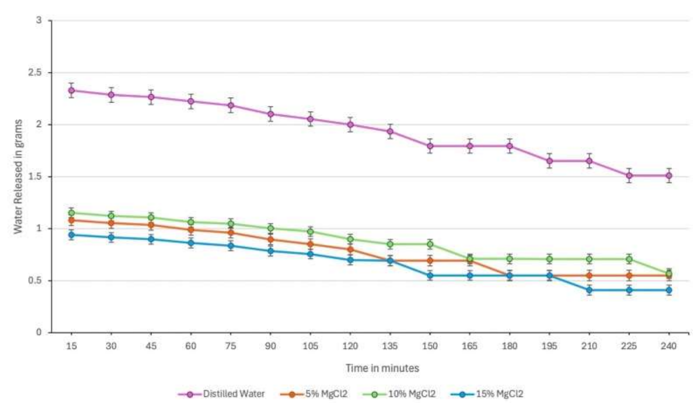

Figure 1.4 shows the amount of water released in grams for the varying concentrations of $MgCl_2$

FINDINGS

From the experiment and tests, we found that across all four experiments, distilled showed the highest water absorption and release rates compared to all the salts. For NaCl experiments, we found that 15% NaCl outperformed the remaining salt concentrations, followed by 5% then 10% concentration. For $MgCl_2$ experiments, 15% and 5% concentrations showed similar trends across tests with 10% lagging behind. This finding proved our hypothesis that salt concentrations reduce water release and absorption rates. However, our second hypothesis that increasing salt concentrations correlate with decreasing absorption rates was disproved since 15% concentrations outperformed 10% in most experiments.

CONCLUSION

The results of this study demonstrate that the presence of dissolved salts significantly decreases the water absorption and release kinetics of hydrogels. However, there was no correlation between increasing concentration of salt and water absorption or release. These findings suggest that ionic contamination from salts impairs the hydrogel's ability to absorb and release water efficiently. Hydrogels, when functioning properly in agricultural settings, should be able to absorb, retain, and release water in the right proportions only when plants need it.

Therefore, if ionic impurities reduce the absorption and release rates, the hydrogels may experience delays in absorbing water, allowing excessive water to seep into the soil for prolonged periods. Similarly, the delayed release of water could deprive plants of crucial moisture during their most critical growth phases, potentially leading to dehydration and hindered growth. Therefore, formulations which resists such conditions should be explored

REFERENCES

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APPENDIX

Tukey HSD results

treatments pair	Tukey HSD Q statistic	Tukey HSD p-value	Tukey HSD inferfence
A vs B	7.4097	0.0010053	** p<0.01
A vs C	10.2995	0.0010053	** p<0.01
A vs D	5.2609	0.0030460	** p<0.01
B vs C	2.8898	0.1879965	insignificant
B vs D	2.1488	0.4363170	insignificant
C vs D	5.0386	0.0047983	** p<0.01

Fig 1.6 shows the Tukey HSD for water absorption for NaCl

Tukey HSD results

treatments pair	Tukey HSD Q statistic	Tukey HSD p-value	Tukey HSD inferfence
A vs B	32.5071	0.0010053	** p<0.01
A vs C	36.1035	0.0010053	** p<0.01
A vs D	34.4474	0.0010053	** p<0.01
B vs C	3.5964	0.0634933	insignificant
B vs D	1.9402	0.5193711	insignificant
C vs D	1.6562	0.6307055	insignificant

Fig 1.7 shows the Tukey HSD for water release $MgCl_2$

Anova: Single Factor (Water Release For MgCl2)

SUMMARY

Groups	Count	Sum	Average	Variance
DISTILLED WATER	16	72.69	4.543125	0.077266517
5% MgCl2	16	44.5043	2.78151875	0.040804079
10% MgCl2	16	41.386	2.586625	0.034282117
15% MgCl2	16	42.822	2.676375	0.03559625

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	41.89199295	3	13.96399765	297.1870125	5.94498E-36	2.758078296
Within Groups	2.819234434	60	0.046987241			
Total	44.71122738	63				

Fig 1.8 shows the ANOVA test for Water release for $MgCl_2$

Tukey HSD results

treatments pair	Tukey HSD Q statistic	Tukey HSD p-value	Tukey HSD inferfence
A vs B	26.0841	0.0010053	** p<0.01
A vs C	33.7395	0.0010053	** p<0.01
A vs D	20.6169	0.0010053	** p<0.01
B vs C	7.6553	0.0010053	** p<0.01
B vs D	5.4672	0.0015303	** p<0.01
C vs D	13.1226	0.0010053	** p<0.01

Fig 1.9 shows the Tukey HSD for water release NaCl

Tukey HSD results

treatments pair	Tukey HSD Q statistic	Tukey HSD p-value	Tukey HSD inferfence
A vs B	8.6968	0.0010053	** p<0.01
A vs C	12.8476	0.0010053	** p<0.01
A vs D	8.9933	0.0010053	** p<0.01
B vs C	4.1508	0.0261894	* p<0.05
B vs D	0.2965	0.8999947	insignificant
C vs D	3.8543	0.0438812	* p<0.05

Fig 1.9 shows the Tukey HSD for water absorption $MgCl_2$

Anova: Single Factor (Water AbsorptionTest For MgCl2)

SUMMARY

Groups	Count	Sum	Average	Variance
DISTILLED WATER	12	113.9	9.491666667	0.129924242
5% MgCl2	12	105.1	8.758333333	0.099015152
10 MgCl2	12	100.9	8.408333333	0.040833333
15% MgCl2	12	104.8	8.733333333	0.071515152

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	7.545625	3	2.515208333	29.47902331	1.31718E-10	2.816465817
Within Groups	3.754166667	44	0.08532197			
Total	11.29979167	47				

Fig 2.1 shows the ANOVA test for Water absorption for $MgCl_2$

Anova: Single Factor (Water Release For NaCl)

SUMMARY

Groups	Count	Sum	Average	Variance
DISTILLED WATER	16	67.211	4.2006875	0.087234363
5% NaCl	16	45.22	2.82625	0.040444067
10 % NaCl	16	38.898	2.431125	0.02111985
15% NaCl	16	49.735	3.1084375	0.034142263

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	27.6317829	3	9.210594292	201.3898988	2.83586E-31	2.758078296
Within Groups	2.74410813	60	0.045735135			
Total	30.375891	63				

Fig 2.1 shows the ANOVA test for Water Release for NaCl

Anova: Single Factor (Water Aborption For NaCl)

SUMMARY

Groups	Count	Sum	Average	Variance
DISTILLED WATER	12	44.8	3.733333333	0.435151515
5% NaCl	12	34.8	2.9	0.090909091
10 % NaCl	12	30.9	2.575	0.014772727
15% NaCl	12	37.7	3.141666667	0.066287879

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	8.614166667	3	2.871388889	18.91806006	4.98477E-08	2.816465817
Within Groups	6.678333333	44	0.151780303			
Total	15.2925	47				

Fig 2.2 shows the ANOVA test for Water Absorption for Nacl

Number of groups: 4

Significance level: 0.05

Group	Sample size	Variance
1	16	0.07726651
2	16	0.04080407
3	16	0.03428211
4	16	0.03559625

Degrees of freedom	Test statistic (T)	P-value
3	3.47181	0.32443

Since the P-value (0.32443) is bigger than the significance level (0.05), we cannot reject the null hypothesis of equal variances across groups.

Fig 2.3 shows the Bartlett test for Water Release for $MgCl_2$

Number of groups: 4

Significance level: 0.05

Group	Sample size	Variance
1	12	0.12992424
2	12	0.09901515
3	12	0.04083333
4	12	0.07151515

Degrees of freedom	Test statistic (T)	P-value
3	3.67372	0.29891

Since the P-value (0.29891) is bigger than the significance level (0.05), we cannot reject the null hypothesis of equal variances across groups.

Fig 2.4 shows the Bartlett test for Water Release for $MgCl_2$